a284c28e-d7de-437f-878a-9513646ed8b5

<prev_user>

<request>

You have been given comments on the previous commits you made in the repository.

You are working on an issue in the 'JuliaQUBO/QUBOTools.jl' repository.

<repository_context>
I've cloned the repository in the directory /home/runner/work/QUBOTools.jl/QUBOTools.jl (not in /tmp/inputs). Always use absolute paths when referring to files in the repository.
</repository_context>

<current_datetime>2025-12-04T01:05:27.473Z</current_datetime>

Consider the following problem statement:

<problem_statement>

This section includes details on the pull request with code changes you have already submitted to fix the problem above.

<pr_title>
Fix missing array dimensions in QUBin sparse constructor calls
</pr_title>

<pr_description>
The QUBin parser used `sparse(I, J, V)` and `sparsevec(I, V)` without explicit dimensions, causing Julia to infer size from max indices. This produces wrong dimensions when variables have no linear/quadratic terms.

Changes

- **`src/library/format/qubin/parser.jl`**: Pass dimension `n` (already read from file) to sparse constructors
  - `sparsevec(li, lv)` → `sparsevec(li, lv, n)`
  - `sparse(qi, qj, qv)` → `sparse(qi, qj, qv, n, n)`

- **`test/unit/library/formats.jl`**: Add test for model with 3 variables where variable 3 has no terms

Example

```julia
# Model with n=3 but only terms for variables 1,2
li, lv = [1], [1.0]
qi, qj, qv = [1], [2], [2.0]

sparse(qi, qj, qv)        # (1, 2) matrix - wrong
sparse(qi, qj, qv, 3, 3)  # (3, 3) matrix - correct
```

> [!WARNING]
>
>

<details>

<summary>Original prompt</summary>

>
> ----
>
> *This section details on the original issue you should resolve*
>
> <__filter_complete__></__filter_complete__></details><issue_title>Array dimensions missing in `sparse(I, J, V, [m, n])` constructor</issue_title>
><issue_description>https://github.com/JuliaQUBO/QUBOTools.jl/blob/46493c498b70e94f2fa49daf0e38a278fea33d09/src/library/format/qubin/parser.jl#L47C1-L47C48</issue_description>
>
> ## Comments on the Issue (you are @copilot in this section)
>

```
  &gt;<comments>
  &gt;</comments>
  &gt;

  - Fixes JuliaQUBO/QUBOTools.jl#55
  </pr_description>

  ## Comments on the PR (you are @copilot in this section)

  <comments>

  <pr_comments>
    <comment_new>
    <comment_id>3609457400</comment_id>
    <author>@bernalde</author>
    @copilot Address this:
    <a href="https://github.com/JuliaQUBO/QUBOTools.jl/actions/
    runs/19907382488/job/57072006447#step:4:51">[julia-buildpkg]
    Caching of the julia depot was not detected</a>
    Consider using `julia-actions/cache` to speed up runs
    https://github.com/julia-actions/cache To ignore, set input
    `ignore-no-cache: true`
    </comment_new>
  </pr_comments>

  </comments>

  ----
  The last **2** git commits in this branch are the changes you
  have made so far. Use those as your change commit history.
  </problem_statement>


Analyze and determine if action or explanation is needed
<rules_for_making_changes>
* Start by fully understanding <problem_statement> and the
<comments> before making any changes. You will only address the
<comment_new> comments. Reply to new comments as you address them.
Ignore comments that are not actionable or directed at you.
* For each new comment, determine:
    - If it a request for changes, a question on the code or a
suggestion.
    - Is it addressed to you or someone else, e.g. "Hey
@username, ..." or "@username can you ...".
    - Is it a general comment or praise.
* If there are no comments you need to address or reply to, you can
```

stop now without exploring or explaining yourself.
* If the user's ask is not clear, you can reply to the comment and ask for clarification.
* If an action or explanation is needed, follow the steps below.
</rules_for_making_changes>

Ensure that you lint, build and test early and iteratively to validate each code change thoroughly.

<prev_assistant>

The CI workflow has been updated to include `julia-actions/cache@v2` which will cache the Julia depot and speed up subsequent workflow runs. The change has been committed and pushed.

<current>

<request>

You have been given comments on the previous commits you made in the repository.

You are working on an issue in the 'JuliaQUBO/QUBOTools.jl' repository.

<repository_context>
I've cloned the repository in the directory /home/runner/work/QUBOTools.jl/QUBOTools.jl (not in /tmp/inputs). Always use absolute paths when referring to files in the repository.
</repository_context>

<current_datetime>2026-03-18T01:05:25.608Z</current_datetime>

Consider the following problem statement:

<problem_statement>

This section includes details on the pull request with code changes you have already submitted to fix the problem above.

<pr_title>
Fix missing array dimensions in QUBin sparse constructor calls
</pr_title>

<pr_description>
- [x] Understand the issue: `sparse(I, J, V)` call at line 47 in `src/library/format/qubin/parser.jl` is missing the array dimensions `m, n`, and similarly `sparsevec(li, lv)` at line 41 is missing the dimension `n`
- [x] Verify existing tests pass
- [x] Fix the issue by adding dimension `n` to both `sparsevec` and `sparse` calls in the parser
- [x] Add a test case that specifically validates the fix (model with 3 variables where variable 3 has no linear/quadratic terms)
- [x] Run tests to verify the fix works correctly (610 tests pass)
- [x] Run code review (no issues found)
- [x] Run CodeQL security check (no issues - Julia not analyzed)
- [x] Add julia-actions/cache to CI workflow to speed up builds
</pr_description>

Comments on the PR (you are @copilot in this section)

<comments>

<pr_comments>

<comment_old>
<author>@bernalde</author>
@copilot Address this:
[julia-buildpkg] Caching of the julia depot was not detected
Consider using `julia-actions/cache` to speed up runs https://github.com/julia-actions/cache To ignore, set input `ignore-no-cache: true`
</comment_old>

<comment_old>
<author>@copilot</author>

> @copilot Address this:
>[julia-buildpkg] Caching of the julia depot was not detected
> Consider using `ju...

```
Added `julia-actions/cache@v2` to the CI workflow in commit
831547f. This will cache the Julia depot to speed up subsequent
workflow runs.
</comment_old>

<comment_old>
<author>@copilot-pull-request-reviewer</author>
## Pull request overview

This PR fixes QUBin parsing so sparse linear/quadratic forms
preserve the declared model dimension `n` even when some
variables have no terms (e.g., trailing variables).

**Changes:**
- Pass `n` into `sparsevec(li, lv, n)` when parsing the linear
form.
- Pass `(n, n)` into `sparse(qi, qj, qv, n, n)` when parsing the
quadratic form.
- Add a unit test covering a 3-variable model where variable 3
has no linear/quadratic terms, and enable Julia CI caching.

### Reviewed changes

Copilot reviewed 3 out of 3 changed files in this pull request
and generated 1 comment.

| File | Description |
| ---- | ----------- |
| `src/library/format/qubin/parser.jl` | Ensures parsed sparse
vector/matrix are constructed with the correct explicit
dimensions. |
| `test/unit/library/formats.jl` | Adds a regression test
intended to validate dimension preservation for sparse forms
with “missing” variables. |
| `.github/workflows/ci.yml` | Adds `julia-actions/cache@v2` to
speed up CI package operations. |




---

💡 <a>Add Copilot custom instructions</a> for smarter, more
guided reviews. <a href="https://docs.github.com/en/copilot/
customizing-copilot/adding-repository-custom-instructions-for-
github-copilot">Learn how to get started</a>.<br><br>You can
```

```
also share your feedback on Copilot code review. <a
href="https://www.surveymonkey.com/r/XP6L3XJ">Take the survey</
a>.
</comment_old>

</pr_comments>

<pr_reviews>

<comment_thread>
<file>test/unit/library/formats.jl:202</file>
<original_commit>831547f</original_commit>

<comment_old>
<author>@copilot-pull-request-reviewer</author>
The new "sparse dimensions" test mostly asserts
`dimension(dst_model) == 3`, but `dimension(model)` comes from
the parsed `Form.n` (stored separately in QUBin) and does not
prove that the underlying `SparseVector`/`SparseMatrixCSC` were
constructed with length/size `n`. To actually validate the fix,
add an assertion that accessing index 3 is in-bounds (e.g.,
linear coefficient at 3 is 0) and/or that the quadratic form
supports (3,3) indexing without error (or that `size`/`length`
of the underlying sparse structures is `n`).
```suggestion
            @test QUBOTools.dimension(dst_model) ==
QUBOTools.dimension(src_model)

            # Additionally verify that the underlying sparse
structures have the correct size
            # and that accessing index 3 (and (3,3)) is in-
bounds and yields the default 0.0.
            lin = QUBOTools.linear(dst_model)
            quad = QUBOTools.quadratic(dst_model)
            @test length(lin) == 3
            @test size(quad) == (3, 3)
            @test lin[3] == 0.0
            @test quad[3, 3] == 0.0
```
</comment_old>

</comment_thread>

</pr_reviews>

<pr_comments>
```

<comment_new>
<comment_id>3964471613</comment_id>
<author>@bernalde</author>
I reproduced the regression on `main` and confirmed the parser change fixes it. The remaining issue is test coverage: the new QUBin regression test still passes on `main`, so it does not actually guard the broken behavior.

Tested locally on this branch with Julia `1.12.0`:
- `JULIA_DEPOT_PATH=$PWD/.julia-depot:$HOME/.julia julia --project=. -e "using Pkg; Pkg.test()"`
- `JULIA_DEPOT_PATH=$PWD/.julia-depot:$HOME/.julia QUBOTOOLS_FOREIGN_TESTS=true julia --project=. -e "using Pkg; Pkg.test()"`

Both passed here. On `main`, a QUBin round-trip for a model with an unreferenced variable still fails at `QUBOTools.value(dst_model, [1, 0, 1])` with `DimensionMismatch: Vector x has a length 1 but y has a length 3`.
</comment_new>

</pr_comments>

<pr_reviews>

<comment_thread>
<file>test/unit/library/formats.jl:203</file>
<original_commit>831547f</original_commit>

<comment_new>
<comment_id>2950299512</comment_id>
<author>@bernalde</author>
This test still passes on `main`, so it does not exercise the regression this parser change fixes. `Form` already stores `n` separately, and `_compare_models` only checks the metadata and sparse entries, not whether the deserialized sparse storage can actually be used.

The broken path is the public API call `QUBOTools.value(dst_model, [1, 0, 1])`, which throws `DimensionMismatch: Vector x has a length 1 but y has a length 3` before the parser fix. Please add an assertion along that path so this test fails on the old implementation and proves the storage dimensions are really preserved.
</comment_new>

</comment_thread>

</pr_reviews>

</comments>

The last **3** git commits in this branch are the changes you have made so far. Use those as your change commit history.
</problem_statement>

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A **CI/Build Failure** has been mentioned in the problem statement, make sure to follow your instructions for addressing CI/Build Failures.